Counterexamples to Yust's Pitch-Space and Pitch-Class Folding Conjectures

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Abstract

We give explicit counterexamples to both arbitrary-cardinality folding characterizations proposed by Yust in 2022. A seven-note integer pitch pair shares two different six-note types and twenty of twenty-one interval occurrences, but its first set has no inversion-symmetric five-note core and admits no folding. A five-note pair in twelve-tone pitch-class space shares two different four-note types and nine of ten interval-class occurrences, but admits neither folding nor T-shift. Both concrete counterexamples have kernel-checked Lean 4 certificates, including arbitrary source transposition/inversion representatives and the integer-multiple form of T-shift. We also prove an infinite pitch-space family in ordinary mathematics. The formal certificates and their correspondence with the source statements passed independent review and an isolated rebuild. Established novelty or publication priority is not claimed.

1 Statements and conventions

Yust's Conjecture 1 characterizes folding of equal-cardinality pitch sets by two shared codimension-one subset types and all but one pairwise interval occurrence. Conjecture 2 proposes the analogous pitch-class characterization, allowing either folding or T-shift [1, p. 8]. The preceding discussion and Proposition 4 supply the relevant conventions: subset types are taken up to transposition and inversion, interval content retains multiplicity, and an n-note folding requires an inversion-symmetric $(n-2)$-note core [1, pp. 4–7].

We work in $G = \mathbb{Z}$ or $G = \mathbb{Z}/12\mathbb{Z}$ and write

$$T_s(x) = s + x, \qquad I_s(x) = s - x.$$

For finite subsets $S, T \subset G$, write $S \sim_{\mathrm{TI}} T$ when $T = T_s(S)$ or $T = I_s(S)$ for some s. In integer pitch space, let

$$\Delta(S) = \biguplus_{\substack{u,v \in S \\ u<v}} \{v - u\}$$

be the multiset of unordered pairwise intervals. In $\mathbb{Z}/12\mathbb{Z}$, use instead the smaller modular distance in $\{1, \dots, 6\}$; its multiplicity vector is denoted $\nu(S) = (\nu_1(S), \dots, \nu_6(S))$. Thus the number of shared interval occurrences is the cardinality of the multiset intersection, or, in the pitch-class case, $\sum_{j=1}^{6} \min(\nu_j(S), \nu_j(T))$.

Definition 1 (Operations used in the source)**.** *Choose distinct $a, b \in S$ and put $C = S \setminus \{a, b\}$. A folding with index s requires $I_s(C) = C$ and replaces S by*

$$(S \setminus \{a\}) \cup \{s - a\}.$$

A T-shift requires $T_x(C) = C$ and, for some $k \in \mathbb{Z}$, replaces S by

$$(S \setminus \{a\}) \cup \{a + kx\}.$$

The output must retain the original cardinality. When comparing types, source and target representatives may be changed by transposition or inversion.

Each operation is a *single* folding or shift, not a sequence of such operations. Each counterexample below has distinct notes, inequivalent whole sets, two *different* shared subset types, and exactly $\binom{n}{2} - 1$ shared interval occurrences. Accordingly, the failures do not depend on repeated-note degeneracies, counting one subset type twice, or interpreting "share" as "at least" rather than "exactly."

2 A seven-note pitch counterexample

Theorem 2. *The integer pitch sets*

$$A = \{0, 1, 2, 3, 6, 7, 9\}, \qquad B = \{0, 1, 2, 3, 6, 8, 9\}$$

share two different six-note types and exactly twenty pairwise interval occurrences. They are not equivalent under transposition or inversion, and no representative of A admits a folding to any target. Thus the forward implication in Yust's Conjecture 1 is false.

Proof. The first common six-set is

$$A \setminus \{7\} = B \setminus \{8\} = \{0, 1, 2, 3, 6, 9\}.$$

For the second, inversion I_9 gives

$$I_9(A \setminus \{2\}) = I_9(\{0, 1, 3, 6, 7, 9\}) = \{0, 2, 3, 6, 8, 9\} = B \setminus \{1\}.$$

The two source six-sets are inequivalent: their common endpoints $0, 9$ force any transposition between them to be T_0 and any inversion to be I_9, and neither maps one to the other.

Their interval multiplicities are

length	1	2	3	4	5	6	7	8	9
$\Delta(A)$	4	3	3	2	2	3	2	1	1
$\Delta(B)$	4	3	3	1	2	3	2	2	1

so, writing $\setminus_m$ for multiset difference,

$$\Delta(A) \setminus_m \Delta(B) = \{4\}, \qquad \Delta(B) \setminus_m \Delta(A) = \{8\}.$$

The shared multiset has cardinality $20 = \binom{7}{2} - 1$. Different interval multisets also exclude $A \sim_{\mathrm{TI}} B$.

An odd finite reflection-invariant set contains the reflection center. A symmetric five-note subset of A would therefore require a center in A and two distinct reflected pairs. The available pairs are

center	reflected pairs in A
$0, 9$	none
1	$\{0, 2\}$
2	$\{1, 3\}$
3	$\{0, 6\}$
6	$\{3, 9\}$
7	none.

No center supports two pairs, so no symmetric five-note core exists. Consequently A admits no seven-note folding. Transposition and inversion preserve the existence and cardinality of symmetric subsets, so the same obstruction holds for every representative of A. □

3 An infinite pitch-space family

Theorem 3. *For every integer $t \geq 6$, the sets*

$$A_t = \{0, 1, 2, 3, t, t+1, t+3\},$$
$$B_t = \{0, 1, 2, 3, t, t+2, t+3\}$$

are counterexamples of the same kind as in Theorem 2. The exchanged interval occurrences are $t-2$ and $t+2$.

Proof. Their common six-set is

$$C_t = A_t \setminus \{t+1\} = B_t \setminus \{t+2\} = \{0, 1, 2, 3, t, t+3\}.$$

Also,

$$I_{t+3}(A_t \setminus \{2\}) = B_t \setminus \{1\}.$$

The source six-sets have endpoints $0, t+3$. Direct comparison excludes both T_0 and I_{t+3} as maps between them, so the shared types are different.

The distances from the exceptional point to C_t are respectively

$$\{|t+1-c| : c \in C_t\} = \{t+1, t, t-1, t-2, 1, 2\},$$
$$\{|t+2-c| : c \in C_t\} = \{t+2, t+1, t, t-1, 2, 1\},$$

where both sides are multisets. All distances within C_t remain unchanged. Hence the full multisets exchange precisely $t-2$ for $t+2$, have twenty shared occurrences, and are different. Thus $A_t \not\sim_{\mathrm{TI}} B_t$.

Suppose a symmetric five-note subset exists. Its center is a member of A_t. A center in the lower block $\{0, 1, 2, 3\}$ supports at most one reflected pair wholly in that block. A pair involving the upper block has sum at least $t \geq 6$, whereas twice the center is at most 6. Equality is possible only for $t = 6$, center 3, and pair $\{0, 6\}$. That center supports no reflected pair wholly in the lower block.

A center in the upper block $\{t, t+1, t+3\}$ has twice its value at least $2t$. A pair involving the lower block has sum at most $t+6$. Equality is possible only for $t = 6$, center 6, and pair $\{3, 9\}$. That center has no pair wholly in the upper block. The upper block itself contains no three-term arithmetic progression. In all cases the center supports at most one pair, contradicting the required five-note symmetry. □

Remark 4 (Formalization boundary). Theorem 3 is an ordinary mathematical proof. The current Lean certificate proves its concrete $t = 6$ instance and the pitch-class example below. It does not assert a formal theorem for the parameterized family. Even-cardinality symmetric subsets are not excluded by the argument; the obstruction is specifically to a five-note core.

4 A five-note pitch-class counterexample

Theorem 5. *In $\mathbb{Z}/12\mathbb{Z}$, let*

$$P = \{0, 1, 3, 4, 6\}, \qquad Q = \{0, 1, 3, 4, 7\}.$$

These sets share two different four-note types and exactly nine interval-class occurrences. They are not equivalent under transposition or inversion and are related by neither folding nor T-shift, even after arbitrary representative changes. Thus the forward implication in Yust's Conjecture 2 is false.

Proof. The first common four-set is

$$P \setminus \{6\} = Q \setminus \{7\} = \{0, 1, 3, 4\}.$$

For the second,

$$I_7(P \setminus \{1\}) = I_7(\{0, 3, 4, 6\}) = \{1, 3, 4, 7\} = Q \setminus \{0\}.$$

These source subset types differ because $\{0, 1, 3, 4\}$ contains two semitone pairs, while $\{0, 3, 4, 6\}$ contains one.

Their interval-class vectors are

$$\nu(P) = (2, 2, 3, 1, 1, 1), \qquad \nu(Q) = (2, 1, 3, 2, 1, 1).$$

They exchange one ic2 for one ic4, so their overlap is $9 = \binom{5}{2} - 1$ and $P \not\sim_{\mathrm{TI}} Q$.

The only inversion-symmetric three-note subset of P is $\{0, 3, 6\}$, with index 6. To check this directly, the center of such an odd subset must be a member $r \in P$ and its other two elements must sum to $2r$ modulo 12. Among the possible centers $0, 1, 3, 4, 6$, only $r = 3$ has such a pair, namely $\{0, 6\}$. Folding either of the remaining notes therefore gives

$$1 \longmapsto 5 : \quad \{0, 3, 4, 5, 6\}, \qquad 4 \longmapsto 2 : \quad \{0, 1, 2, 3, 6\}.$$

Both outputs have vector $(3, 2, 2, 1, 1, 1)$, unlike Q, so neither reaches its type.

A nonidentity translation stabilizing a three-element subset has order three: its equal-sized translation orbits partition that subset. Such a subset is necessarily an augmented triad $\{r, r+4, r+8\}$. The set P contains no augmented triad, so none of its three-note cores has a nonidentity translational symmetry. The identity shift leaves P unchanged, which is not Q's type.

Finally, stabilizing translations form a subgroup. Every integer multiple of a stabilizing translation still stabilizes the same core, so testing all stabilizing translations includes the complete definition of T-shift. Transposition/inversion changes of source representatives conjugate these operations to operations of the same form and preserve core symmetries. The obstruction consequently applies to every representative of P. □

5 Formal certificate and independent verification

The file `formal/Counterexamples.lean` models pitch sets by `Finset` $\mathbb{Z}$ and pitch-class sets by `Finset (Fin 12)`. It defines unordered pairwise interval multisets directly and retains their multiplicities. The main exported theorems are

`YustFolding.conjecture_one_false`, `YustFolding.conjecture_two_false`.

The formal conjecture statements include $n \geq 3$, inequivalent whole sets, two distinct shared deletion types, exact interval overlap, arbitrary source representatives, and the printed integer-multiple T-shift predicate. These are stricter content-side hypotheses than the printed conjectures explicitly state; the counterexamples satisfy them all.

Unbounded pitch space. If a nonempty subset $C \subset [0, 9] \cap \mathbb{Z}$ satisfies $I_s(C) = C$, then for $x \in C$ we have $s - x \in C$, and hence $s = x + (s - x) \in [0, 18]$. The Lean proof establishes this bound before checking all possible integer indices. It does not replace an unbounded pitch universe by a bounded one. Pullback lemmas transport the obstruction to every transposition and inversion of A: the pulled-back inversion indices are respectively $s - 2u$ and $2u - s$ for a source change by T_u or I_u.

Pitch-class representatives and shifts. Every transposed and inverted source is checked; the operation predicates already permit all target T/I representatives. The stabilizing translations are explicitly packaged as an additive subgroup, and closure under integer multiplication proves that the exact T-shift relation of Definition 4 is included in the checked relation.

Kernel and audit. All finite certificates use ordinary kernel-checked `decide`. There is no admitted proof, new axiom, `native_decide`, or external solver oracle. The exported theorem dependencies are only the standard foundations `propext`, `Classical.choice`, and `Quot.sound`. Python enumeration supported discovery and independent numerical checks but is not imported as a proof dependency.

The frozen source passed an independent semantic review and a fresh isolated Lean compilation, with exit status 0 in 74.84 seconds. No project object files were reused. The audit checked the exact exported statements and the source-to-formal bridges, including multiplicities, distinct subset types, representatives, unbounded axes, and integer-multiple shifts. It found no remaining semantic or proof gap. Receipts and exported statements are included in the accompanying archive. The infinite family was independently reviewed as an ordinary mathematical proof, with its formalization boundary retained.

6 Literature check and limitations

The author-hosted revised paper and the author's publication list [1, 2], containing entries through 2026, were checked on 14 October 2026. Targeted exact-title searches combined with "proof" and "counterexample," and searches for the two named conjectures, found no later primary-source resolution. The Springer chapter landing page and the two linked presentation files were not readable through the available retrieval route. This is a bounded literature check, not proof of originality or absence of prior work.

The results refute the arbitrary-cardinality characterizations. They do not refute the proved pitch-space tetrachordal proposition, classify all exceptional pairs, establish the smallest pitch-space counterexample, or prove a repaired characterization. No publication-priority claim is made.

A Reproducibility

The toolchain is Lean 4.32.2, with mathlib pinned to

`905b95818eb32af7874a58b427f50c1711a5e96c`.

The SHA256 of the audited one-module formal source is

`a884ebb6566c9920f3890c3b9b331a84c7e4a168404482847992a59b92217ace`.

The archive includes `lean-toolchain`, `lakefile.toml`, the full formal source, build logs, independent receipts, enumeration code and results, this LaTeX source, and the compiled paper. On a machine with Elan/Lake, the standard build sequence is

```
lake update
lake exe cache get
lake build
```

The supplied `formal/check.sh` additionally records the command used with the existing research toolchain. The paper can be rebuilt with `build-paper.sh`; its local writable TeX format avoids changing system configuration.

References

[1] Jason Yust. Tetrachordal Folding Operations. In *Mathematics and Computation in Music*, MCM 2022, pp. 292–303. Springer, 2022. DOI: 10.1007/978-3-031-07015-0_24. Author-hosted revised paper, especially pp. 4–8.

[2] Jason Yust. Publications. Boston University. Accessed 14 October 2026.